Ch-4

$$\pi = \left| \frac{d(g(x))}{dx} \right| \qquad g(x_*) = x_*$$

Newton Method

$$g(x) = x - \frac{f(x)}{f'(x)}$$ → find g'

Newton

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\text{Relative error} = \frac{|x_{new} - x_{old}|}{|x_{new}|}$$

Quasi Newton / Secant

$$x_{k+1} = x_k - \frac{f(x_k)\cdot(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$

Plain Rule

Ch - 5

triangular

$$Ax = b$$

$$A = x^{-1} b$$

lower triangle

$$\begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ & & & \end{bmatrix}$$

$$x_i = \frac{b_i - \sum_{k=1}^{i-1} l_{ik} x_k}{l_{ik}} \quad i = 1 \text{ to } n$$

up

$$x_i = \frac{b_i - \sum_{k=i+1}^{n} u_{ik} x_k}{u_{ik}}$$

L U

Step

① $A^{(1)} = A$

② $F^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{bmatrix}$

$m_{21} = \frac{a_{21}}{a_{11}}$

$m_{31} = \frac{a_{31}}{a_{11}}$

③ $A^{(2)} = F^{(1)} \times A^{(1)}$

④ $F^{(2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -m_{32} & 1 \end{bmatrix}$

$m_{32} = \frac{a_{32}}{a_{22}}$

⑤ $A^{(3)} = F^{(2)} \times A^{(2)} = [U]$

⑥ $L = (F^{(1)})^{-1} \times [F^{(2)}]^{-1}$

CamScanner

(7) $[L] \times \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \neq \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$

(8) $[U] \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

CamScanner

Chap - 6

Least square approx

$$Ax = b$$

$$\Rightarrow A^T A x = A^T b$$

orthogonality

(i) matrix notation $= x^T . y$

(ii) vector notation $= \vec{x} . \vec{y}$

inner product

$\vec{x} . \vec{x}$ or $x^T . x$

orthogonal

$x^T . y = 0$

orthonormal

$\vec{x} . \vec{y} = 1$

$\vec{x}^T . x = 1$

$y^T . y = 1$

$Q . Q^T = I$

$Q^T . Q = I$

$Q^{-1} = Q^T$

Q R

u_1, u_2, u_3

↓ Gram Schmidt process

p_1 p_2 p_3

~~xxx~~ normalized

↓

q_1, q_2, q_3 → orthonormal

$$q_k = \frac{p_k}{|p_k|} \qquad p_k = u_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$A = QR \qquad A = \begin{bmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{bmatrix}$$

$$Q = \begin{bmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{bmatrix}$$

$$p_1 = u_1, \quad q_1 = \frac{p_1}{|p_1|}$$

$$p_2 = u_2 - (u_2^T q_1) q_1, \quad q_2 = \frac{p_2}{|p_2|}$$

$$p_3 = u_3 - (u_3^T q_1) q_1 - (u_3^T q_2) q_2$$

CamScanner

$$q_3 = \frac{p_3}{|p_3|}$$

$$R = \begin{bmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_2 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{bmatrix}$$

$$Rx = Q^T b$$

Ch- 7

~~Triangle~~ :

Trapezoidal

$$I_T(f) = \frac{b-a}{2}\left[f(a) + f(b)\right]$$

upper bound error

$$|f(x) - P_n(x)| \le \left|\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n)\right|$$

for integration, $\forall \xi$

$$|I - I_n| \le \left|\frac{f^{(n+1)}(\xi)}{(n+1)!}\right| \int_{max}^{b} |(x-x_0)(x-x_1)\cdots(x-x_n)|\,dx$$

Error :

$$R.E = \frac{|Act - num|}{|Act|}$$

$$\text{percen. E.} = \frac{|Act - num|}{Act|} \times 100\%$$

$$R.P.E. = \frac{|Act - numr|}{|Act|} \times 100\%$$

$$Az \therefore E = |act - nume|$$

Newton cotes

Actual and $I(z) = \int_a^z z(x)\,dx$

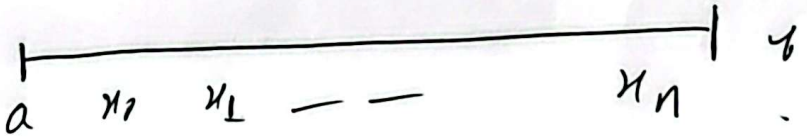
numerical Int, $I_n(z) = \int_a^z p_n(x)\,dx$

$$= \int_a^z \sum_{k=0}^{n} \lambda_k(x)\, z(x_k)\,dx$$

$$= \sum_{k=0}^{n} z(x_k) \int_a^z \lambda_k(x)\,dx$$

closed

$$h = \frac{z-a}{n}$$



open

$$h = \frac{z-a}{n+2}$$

CamScanner

Simpsons Rule

$$\text{Trape Rule} = \int_a^b P_1(x)\,dx \qquad n=1$$

$$\text{Simpson's Rule} = \int_a^b P_2(x)\,dx \qquad n=2$$

$$I_2(f) = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right]$$

Com Newton – Cotes

$$C_{1,m}(f) = \frac{h}{2}\left[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots \right.$$

$$\left. \cdots + 2f(x_{m-1}) + f(x_m)\right]$$